SentinelAI — AI-Powered Phishing and Cyber Threat Analyzer

Detect Warning Signs. Understand Risks. Stay Safer Online.

SentinelAI is a cybersecurity mini project designed to help users identify possible phishing attempts by analyzing suspicious URLs and text messages. It highlights potential warning signs, provides a risk assessment, and recommends safer online practices.

Project Type: BTech CSE Mini Project
Domain: Cybersecurity, Web Development, AI Concepts
Project Status: Student Prototype
Primary Goal: Improve awareness of phishing attacks and suspicious online messages.

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1. Introduction

The internet is widely used for online banking, shopping, education, communication, and digital payments. However, cybercriminals may use phishing links, fake websites, and fraudulent messages to trick people into revealing personal information.

Many users find it difficult to recognize these threats.

SentinelAI aims to make basic phishing awareness easier by analyzing user-submitted URLs and message text, identifying suspicious indicators, and displaying useful safety recommendations.

2. Problem Statement

Phishing attacks often use fake messages and misleading websites to steal sensitive information such as login credentials, banking details, and personal data.

Users may not recognize warning signs such as suspicious domains, urgent language, unexpected prizes, or requests for payments.

Therefore, a simple and user-friendly tool is needed to help users examine suspicious links and messages and understand the risks before taking action.

3. Objectives

The main objectives of SentinelAI are:

- To help users recognize possible phishing URLs.

- To analyze suspicious SMS and email text.
- To identify common phishing warning signs.
- To display an understandable risk assessment.
- To provide safety recommendations.
- To maintain a history of previous analyses, where implemented.
- To promote cybersecurity awareness among students and everyday internet users.

4. Proposed Solution

SentinelAI provides a web-based interface where users can submit a suspicious URL or paste a message for analysis.

The application examines available indicators and presents a risk category, detected warning signs, and recommended precautions.

The initial version can use rule-based analysis. Machine learning and external threat-intelligence services are future enhancements unless they have actually been integrated into the application.

5. Key Features

5.1 Dashboard

Provides an overview of the application and its available analysis tools. Any statistics displayed should reflect actual scan records or be clearly identified as sample data.

5.2 URL Analyzer

Allows users to enter a URL and checks for configured warning signs, such as:

- Unusual URL structure.
- Use of IP addresses instead of familiar domain names.
- Suspicious domain patterns.
- Misleading or excessive subdomains.
- Other configured URL indicators.

5.3 Message Analyzer

Allows users to paste suspicious SMS or email content and checks for warning signs, such as:

- Urgent requests for action.
- Unexpected prize or reward claims.
- Requests for payment or sensitive information.
- Suspicious account-verification messages.
- Pressure to click a link immediately.

5.4 Risk Assessment

Displays a risk score or category based on the application's implemented detection rules. Results should explain which indicators contributed to the assessment.

Important: A heuristic risk score is not a verified probability that a message or URL is malicious.

5.5 Safety Recommendations

Provides suggestions such as:

- Do not click unexpected links.
- Never share passwords, PINs, or OTPs.
- Verify messages through official websites or contact channels.
- Report suspected phishing attempts to the relevant service.

5.6 Scan History

Can display previously analyzed items and their results, provided history storage has been implemented and enabled.

6. How the System Works

1. The user opens SentinelAI.
2. The user selects URL analysis or message analysis.
3. The user enters a URL or pastes message text.
4. The application evaluates the input using its configured detection logic.
5. The application identifies possible warning signs.
6. A risk assessment and safety recommendations are displayed.
7. The result may be saved in scan history if that feature is enabled.

Workflow:

```
User Input → Analysis → Warning Sign Detection → Risk Assessment → Safety Recommendations
```

7. Technology Stack

The final technology list should match the actual implementation generated by Base44.

Technology	Purpose
Base44	AI-assisted application building and development platform

HTML, CSS and JavaScript or generated frontend framework	User interface and application interactions
JavaScript-based analysis rules or implemented backend logic	Basic phishing indicator analysis
Base44 database or browser storage	Data and scan history, depending on implementation
Machine learning libraries	Optional future enhancement, not assumed to be implemented

8. Example Use Cases

Example 1: Fake Bank KYC Message

A user receives a message claiming that their bank account will be blocked unless they update their KYC details through an unfamiliar link.

Possible warning signs: Urgent language, an unexpected account warning, and a request to use an unfamiliar link.

Recommendation: Open the bank's official website independently or contact the bank through a verified channel.

Example 2: Unexpected Prize Message

A message claims that the user has won a prize but must pay a fee to receive it.

Possible warning signs: An unexpected reward and a request for payment.

Recommendation: Do not pay the fee or share personal information without independently verifying the claim.

Example 3: Suspicious URL

A user enters a URL containing unusual domain patterns.

Possible warning signs: Suspicious URL structure or other indicators recognized by the application's rules.

Recommendation: Verify the website's identity before entering any sensitive information.

These examples demonstrate possible analysis scenarios; they do not represent verified scan results.

9. Benefits

- Encourages safer internet usage.
- Helps users understand common phishing techniques.
- Provides an accessible interface for basic checks.
- Supports cybersecurity education.
- Demonstrates practical web application development.
- Provides a foundation for future AI-assisted security features.

10. Limitations

- Rule-based checks cannot detect every phishing attack.
- A suspicious-looking URL may be legitimate, and a convincing phishing URL may not trigger any configured rule.
- The application should not claim that a URL is completely safe based only on its score.
- Real-time threat intelligence is unavailable unless an external service has been integrated.
- Machine learning is not part of the implemented system unless a trained model has actually been connected.
- Analysis results should be treated as guidance, not as a guarantee of security.

11. Future Enhancements

Possible future improvements include:

- Integration with trusted threat-intelligence services.
- A trained machine learning model for phishing detection.
- Email security analysis.
- Browser extension support.
- Multilingual phishing-message analysis.
- User authentication and role-based access control.
- Improved reporting and analysis dashboards.
- Additional security testing and performance improvements.

These enhancements are proposed future work, not confirmed existing features.

12. Running the Project

Using the Base44-hosted application

1. Open the deployed SentinelAI application.
2. Navigate to the URL Analyzer or Message Analyzer.
3. Enter a test URL or paste sample message text.
4. Run the analysis.
5. Review the displayed warning signs and recommendations.
6. Open Scan History to check saved results if history is implemented.

Running exported source code

If the project is exported from Base44, follow the setup instructions for the actual generated framework. Add the correct installation commands, environment variables, and start command after confirming the exported project files.

Do not add unverified commands or publish API keys and secrets in the repository.

13. Security and Privacy

- Do not enter real passwords, OTPs, bank details, or other sensitive information into test messages.
- Do not automatically open or visit submitted URLs as part of analysis.
- Avoid storing submitted content longer than necessary.
- Protect user records and credentials if authentication or database storage is implemented.
- Do not publish private API keys or access tokens.
- Use sample data for demonstrations whenever possible.

14. Project Structure

The following is a conceptual structure. Replace it with the actual files and folders in your exported project.

```
SentinelAI/
├── README.md
├── frontend/             # If generated in the exported project
├── backend/              # If a separate backend exists
├── assets/               # Images, icons, and other assets
└── documentation/        # Project report and screenshots
```

15. Conclusion

SentinelAI is a student cybersecurity project focused on helping users recognize possible phishing links and suspicious messages. By combining a simple interface, warning-sign analysis, risk assessments, and safety recommendations, it demonstrates how software can support cybersecurity awareness.

The project also provides a foundation for future enhancements using machine learning and trusted threat-intelligence services.

SentinelAI promotes informed decisions and safer online behavior. It is an educational tool, not a replacement for professional security software.

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Project: SentinelAI — AI-Powered Phishing and Cyber Threat Analyzer
GitHub Repository: [Add your repository URL here]

Disclaimer

This project is intended for educational and cybersecurity-awareness purposes only. Its analysis results are not a guarantee that a URL or message is safe or malicious. Always verify suspicious communications through official channels.